

Final Project 2025.

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Weighting: 40% of final course grade

Due date/time: Thursday 23rd October at 23:59

Regarding A.I. Usage:

In this assessment task, you may use AI tools to generate materials or content related to the task. **To gain credit the AI tools must only reflect the methods taught in the course Modules.** You are expected to complete the assignment based on the examples and code covered in the lecture materials and practicals. **If you use AI to generate code, there is a high chance that the process will not align with the assignment requirements and may produce incorrect solutions. This can significantly lower your marks.** I strongly recommend that you work through the assignment independently, from start to finish, using the lecture and practical materials as your main guide. AI tools should only be used as a support to clarify concepts already covered in class—not as the primary source of code or answers. Also **always reference content generated by AI - if you don't, it's considered plagiarism.**

Tasks & Dataset files:

In this Project, you have **3 Tasks**. You will work **independently** on this Project.

Please find your tasks and data **HERE** in CANVAS, and ensure you work individually on these tasks and data.

Your submission pdf must be in official **written report format**.

Total Marks = 330.

Task 1 (worth 100 marks)

In a study of potential effects of both climate and pollution on disease specific mortality between the years 2010-2020 a team of researchers studied the disease specific averaged weekly mortality in Paris, France and the city's local climate (temperature degrees Fahrenheit), size of pollutants and levels of noxious chemical emissions from cars and industry in the air - all measured at the same points between 2010-2020.

All 5 series i.e. mortality, temperature, pollutants particle size and two chemical emissions (chem1, chem2) between 2010-2020 (508 time points) are given here in [mort.csv](#) .

You will use this data for the calculation of 4 weeks ahead forecasts for mortality.

Your task is to give best 4 weeks ahead forecasts in terms of R squared, AIC, BIC, MASE etc (as is appropriate) for the mortality series. Provide the point forecasts and confidence intervals and corresponding plot for the most optimal model for each method used (DLM, ARDL, polyck, koyck, dynamic, exponential smoothing and state-space model).

Multiple predictors are to be modelled i.e., use more than 1 predictor in regression type models, (i.e. multivariate). Point forecasts and confidence intervals should be reported along with appropriate graphs. Percentiles method for relevant covariates for the forecasts can be used.

Hint: Use MASE () function from the dLagM package to compute MASE for time series regression methods for model comparisons.

Task 2 (worth 100 marks)

In a study of 81 species of Australian plants Hudson & Keatley (2021) investigated whether the day of occurrence of a species first flowering (first flowering day, FFD, a number between 1 -365) is impacted by climate factors such as rainfall (rain),

temperature (temp), radiation level (rad), and relative humidity (RH). The study by Hudson & Keatley essentially explores the influence of long-term climate on the FFD of 81 species of plants from 1984 to 2014.

Your data focuses on one species (of the 81) and contains 5 time series, the FFD time series of the given plant species and the contemporaneous yearly averaged climate variables measured from 1984 – 2014 (31 years). All series are available here in [FFD .csv](#) ↓

Your task is to model FFD and forecast FFD. Single climate predictors (univariate models) are to be tested. Your task is to give best FFD 4 years ahead forecasts for the FFD series. Point forecasts and confidence intervals are required for the forecasts with appropriate graphs.

For forecasting, the future covariates are given here in [Covariate x-values for Task 2 .csv](#) ↓

Provide the point forecasts and confidence intervals and corresponding plot for the most optimal model for **each method used** (DLM, ARDL, polych, koyck, dynamic, exponential smoothing and state-space model).

Use MASE() function from the dLagM package to compute MASE for time series regression methods for model comparisons. Choice of optimal models within a specific method can be assessed from values of R squared, AIC, BIC, MASE etc (as is appropriate to the method).

Hints:

- Perform univariate analyse of predictors.
- Run models with and without intercept.
- Interpret the lag effects – note positive lag coefficient means that flowering is later and negative ones earlier flowering (lower FFD).
- Intercept denotes the intercept mean FFD. You are to analyse one given species (on Canvas).
- Plots are essential ACF PACF, forecasts etc.
- Percentiles method for relevant covariates for the forecasts can be used.

Task 3 (worth 130 marks)

Climate change has become a significant issue in Australia. Australia's extreme climate is vulnerable to climate shifts and has experienced many extreme weather events recently. Research has shown that climate change is influencing flowering timing and the relative flowering orders of plants. Changes in flowering order can have detrimental effects on plant growth and food availability. Little research has examined how flowering orders have changed over time. This data from Hudson & Keatley (2021) explores the influence of long-term climate shifts in Victoria on the relative flowering order similarity of 81 species of plants from 1983 to 2014. The species were ranked annually by the time taken to flower (FFD), and changes in flowering order were measured by computing the similarity between annual flowering order and the flowering order of 1983 using the Rank-based Order similarity metric (RBO). The earliest flowering species is ranked 1 and latest ranked 81 for the given year under study.

A decrease in similarity over time suggests that flowering orders are becoming more dissimilar from their original ordering in 1983. The aim of your analysis is to see if it is possible to determine how flowering orders respond to specific changes in climate by understanding the relationship between flowering order (RBO) and climate conditions. Climate variables are yearly averaged temperature, rainfall, radiation and relative humidity. These relationships may be used – in theory – to forecast future flowering orders. Note the term "flowering order similarity" refers to the similarity between the flowering order of the 81 plant species for a particular year and the flowering order of the same plants in the so-called baseline reference year, 1983 (the first year in the data collection period with enough observations) measured by a rank similarity metric, RBO. RBO values are therefore numbers between 0 and 1. Higher RBO values indicate higher similarity of the order of the first flowering occurrence (based on FFD) of the 81 species from 1983 compared to each of the subsequent years, 1984 to 2014, so the time series are of length 31.

Models will be built using RBO similarity values for the flowering orders of consecutive years as the dependent time series (Y_t). Your data contains 5 time series, the RBO time series of the 81 plant species studied by Hudson & Keatley (2021) and the yearly averaged climate variables measured from 1984 – 2014, as 1983 is the reference year for flowering order FFD ranks. The time series are thereby of length 31, as the RBO metrics utilise 1983 rank orders of FFD as the baseline.

All series are available here in [RBO .csv](#) ↓

For forecasting, the future covariates are given here in [Covariate x-values for Task 3 .csv](#) ↓

Task 3 Part (a): Carry out your analysis based on **univariate** climate regressors (model **one** climate indicator at a time, i.e., **univariate** regressor).

- Modelling methods to try (DLM, ARDL, polyck, koyck, dynlm).
- Choice of optimal models within EACH a specific method can be assessed from values of R squared, AIC, BIC, MASE etc (as is appropriate to the method).

The goal is to forecast RBO **three** years ahead using each regressor one at a time and (use percentiles for the regressors) in forecasting for each of the **best** models within the methods utilised.

Point forecasts and confidence intervals are required to be obtained and reports for the forecasts. Percentiles method for relevant covariates for the forecasts can be used.

Task 3 Part (b): Flowering orders became more dissimilar over the most recent decades, particularly during the Millennium Drought (1997 – 2009), suggesting that flora in Australia is responding to changes in their environment. According to the BoM the drought period for Australia occurred from 1996 to 2009 How would you **now** accommodate for this in your analysis of the Rank RBO. Perform the appropriate analysis and obtain the 3 year ahead forecasts (suggest using the dynlm package) only for part (b)).

Note: You must complete all elements listed in the rubric when working on the tasks. To receive credit, you **must use methods or functions covered in course Modules**. Please be aware that **if you use any method not covered in the course content, you will not receive credit** for that part.

Written report format:

Please see the following webpage to get more information about the required **written report format**:

<https://learninglab.rmit.edu.au/assessments/reports/> ➞

Choice regarding Microsoft Word or R Markdown:

In regards to creating your pdf submission, you are free to simply write your R code, run it, save your output, and then combine both R code and output with your interpretations in a Word document, and then you must save that as a **single pdf file**.

Or alternatively, you can invest time into learning R Markdown coding, but you will need to save your R Markdown output as a pdf (do not submit a .Rmd file in Canvas). Many of you are familiar with R Markdown, but for those of you who are not, here are official R Markdown learning resources if you are interested:

<https://rmarkdown.rstudio.com/> ➞

<https://rmarkdown.rstudio.com/lesson-1.html> ➞

Submission Instructions:

- All reports must be **submitted via the Canvas upload links on this assignment page**.
- Your submission should be **a SINGLE PDF file only**.
- Your report should meet with English language requirements.
- **Late submissions** will be penalised in accordance with the late submission policy explained in the "Course Overview" module pages.

Plagiarism and Academic Misconduct:

Please keep in mind that this is an individual assignment and you should demonstrate your own effort and understanding. If cheating is detected, both the copier and the student copied from will be responsible. Therefore, it is your responsibility to ensure you do not copy or do not allow other classmates to copy your work. Please read through the [academic integrity pages of RMIT University](#) ➞. It is important to understand proper referencing the information that you use in your writings. Be aware that plagiarism and academic misconduct are serious offenses. Please review the [RMIT University policy](#) ➞ on plagiarism.

Required Components for each Task:

Dot-pointed below are the main components of each Task:

- Reporting & Interpretation
- Summary Tables of models
- R codes
- Descriptive Analysis
- Choice of variables/model
- Implementation of models including Forecasting
- Diagnostic checking
- Graphics
- Conclusion

Also see the detailed Marking Rubric at the bottom of this page.

Criteria	Ratings							Pts
Task 1: Reporting & Interpretation	20 Pts >16 to 20 Achieves High distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences in a way to provide a mature discussion of analysis results. There is no missing section. There are no language and formatting issues.	16 Pts >14 to 16 Achieves Distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences. There is no missing section. There are no language and formatting issues.	14 Pts >12 to 14 Meets expectation All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	12 Pts >8 to 12 Pass All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	8 Pts >4 to 8 Needs improvement Inferences are addressed by relevant outputs and given under relevant sections. There are missing sections. There are minor languages and formatting issues.	4 Pts >0 to 4 Poor Inferences are not addressed by relevant outputs and given under relevant sections. Some of the output remains without being inferred. There are language and formatting issues.	0 Pts No marks Lacks the basic elements of a precise report. Some of the outputs have not been associated with inferences or there are inferences without supporting evidence.	20 pts
Task 1: R codes	25 Pts >20 to 25 Achieves High distinction Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The scripts include user-created functions to implement all analyses in a	20 Pts >18 to 20 Achieves Distinction Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The scripts include user-created functions to implement all analyses without using	18 Pts >15 to 18 Meets expectation Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The script is arranged in a way not to repeat the same code chunks.	15 Pts >10 to 15 Pass Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code.	10 Pts >5 to 10 Needs improvement All scripts to run analysis are working properly and suitable explanations are given on the code.	5 Pts >0 to 5 Poor All scripts to run analyses are working properly. However, it is hard to follow codes to identify what each line does.	0 Pts No marks Some of the scripts to run analyses are generating error messages. It is unclear what each line does in the R codes.	25 pts

Criteria	Ratings							Pts
	dynamical way.	the same code chunks repeatedly.						
Task 1: Descriptive analysis	10 Pts >8 to 10 Achieves High distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to demonstrate an outstanding understanding of concepts targeted by the descriptive statistic/plot.	8 Pts >7 to 8 Achieves Distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to shed light on further analyses.	7 Pts >6 to 7 Meets expectation Uses all suitable plots and descriptive statistics and each plot and the descriptive statistic have a suitable interpretation.	6 Pts >4 to 6 Pass Uses all suitable plots and descriptive statistics and some plots and descriptive statistic have a suitable interpretation in one of the tasks.	4 Pts >2 to 4 Needs improvement Some descriptive plots/statistics are used and/or presented all plots/statistics have interpretations.	2 Pts >0 to 2 Poor Some descriptive plots/statistics are used and/or presented plots/statistics have interpretations.	0 Pts No marks Nearly no descriptive analysis has been done and/or the descriptive analysis is insufficient.	10 pts
Task 1: Choice of variables/model	15 Pts >12 to 15 Achieves High distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. Insightful reasons given about the selection of the model.	12 Pts >11 to 12 Achieves Distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is articulated based on the results of descriptive statistics.	11 Pts >9 to 11 Meets expectation A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is explained.	9 Pts >6 to 9 Pass A suitable model is chosen, and all variables are suitable for the aim of analysis.	6 Pts >3 to 6 Needs improvement A suitable model is chosen but some variables in a model are unsuitable for the aim of analysis.	3 Pts >0 to 3 Poor A suitable model is chosen but some or all variables in a model are unsuitable for the aim of analysis.	0 Pts No marks An unsuitable model is chosen or all variables in a model are unsuitable for the aim of analysis.	15 pts
Task 1: Implementation of models including Forecasting	15 Pts >12 to 15 Achieves High distinction All possible models are fitted, and all the relevant outputs are clearly	12 Pts >11 to 12 Achieves Distinction All possible models are fitted, and all the relevant outputs are included in the	11 Pts >9 to 11 Meets expectation All possible models are fitted and there is some effort to draw inferences	9 Pts >6 to 9 Pass All possible models are fitted but there is a limited amount of effort to draw inference from the	6 Pts >3 to 6 Needs improvement Some models fitted but there is a limited amount of effort to draw inference	3 Pts >0 to 3 Poor Some models fitted but there is no attempt to draw inferences from the	0 Pts No marks Specified models are not fitted and there is no attempt to move forward with the analysis.	15 pts

Criteria	Ratings							Pts
	interpreted in a way to inform the next step of the analysis.	comments made on the fitted models.	from the model outputs.	model outputs.	from the model outputs.	model outputs.		
Task 1: Diagnostic checking	15 Pts >12 to 15 Achieves High distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. A mature discussion of the information gathered by diagnostic checking is given.	12 Pts >11 to 12 Achieves Distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. The results are articulated in a way to demonstrate a good understanding of diagnostic checking.	11 Pts >9 to 11 Meets expectation All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	9 Pts >6 to 9 Pass Some aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	6 Pts >3 to 6 Needs improvement Some aspects of diagnostic checking are applied and most of the assumptions of applied approaches are validated.	3 Pts >0 to 3 Poor Some aspects of diagnostic checking are applied and some of the assumptions of applied approaches are validated.	0 Pts No marks There is no effort for diagnostic checking and validation of the assumptions of applied approaches.	15 pts
Task 2: Reporting & Interpretation	20 Pts >16 to 20 Achieves High distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences in a way to provide a mature discussion of analysis results. There is no missing section. There are no language and	16 Pts >14 to 16 Achieves Distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences. There is no missing section. There are no language and formatting issues.	14 Pts >12 to 14 Meets expectation All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	12 Pts >8 to 12 Pass All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	8 Pts >4 to 8 Needs improvement Inferences are addressed by relevant outputs and given under relevant sections. There are missing sections. There are minor languages and formatting issues.	4 Pts >0 to 4 Poor Inferences are not addressed by relevant outputs and given under relevant sections. Some of the output remains without being inferred. There are language and formatting issues.	0 Pts No marks Lacks the basic elements of a precise report. Some of the outputs have not been associated with inferences or there are inferences without supporting evidence.	20 pts

Criteria		Ratings						Pts
	formatting issues.							
Task 2: R codes	25 Pts >20 to 25 Achieves High distinction Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The scripts include user-created functions to implement all analyses in a dynamical way.	20 Pts >18 to 20 Achieves Distinction Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The scripts include user-created functions to implement all analyses without using the same code chunks repeatedly.	18 Pts >15 to 18 Meets expectation Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The script is arranged in a way not to repeat the same code chunks.	15 Pts >10 to 15 Pass Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code.	10 Pts >5 to 10 Needs improvement All scripts to run analysis are working properly and suitable explanations are given on the code.	5 Pts >0 to 5 Poor All scripts to run analyses are working properly. However, it is hard to follow codes to identify what each line does.	0 Pts No marks Some of the scripts to run analyses are generating error messages. It is unclear what each line does in the R codes.	25 pts
Task 2: Descriptive analysis	10 Pts >8 to 10 Achieves High distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to demonstrate an outstanding understanding of concepts targeted by the descriptive statistic/plot.	8 Pts >7 to 8 Achieves Distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to shed light on further analyses.	7 Pts >6 to 7 Meets expectation Uses all suitable plots and descriptive statistics and each plot and the descriptive statistic have a suitable interpretation.	6 Pts >4 to 6 Pass Uses all suitable plots and descriptive statistics and some plots and descriptive statistic have a suitable interpretation in one of the tasks.	4 Pts >2 to 4 Needs improvement Some descriptive plots/statistics are used and/or presented all plots/statistics have interpretations.	2 Pts >0 to 2 Poor Some descriptive plots/statistics are used and/or presented plots/statistics have interpretations.	0 Pts No marks Nearly no descriptive analysis has been done and/or the descriptive analysis is insufficient.	10 pts

Criteria	Ratings							Pts
Task 2: Choice of variables/model	15 Pts >12 to 15 Achieves High distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. Insightful reasons given about the selection of the model.	12 Pts >11 to 12 Achieves Distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is articulated based on the results of descriptive statistics.	11 Pts >9 to 11 Meets expectation A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is explained.	9 Pts >6 to 9 Pass A suitable model is chosen, and all variables are suitable for the aim of analysis.	6 Pts >3 to 6 Needs improvement A suitable model is chosen but some variables in a model are unsuitable for the aim of analysis.	3 Pts >0 to 3 Poor A suitable model is chosen but some or all variables in a model are unsuitable for the aim of analysis.	0 Pts No marks An unsuitable model is chosen or all variables in a model are unsuitable for the aim of analysis.	15 pts
Task 2: Implementation of models including Forecasting	15 Pts >12 to 15 Achieves High distinction All possible models are fitted, and all the relevant outputs are clearly interpreted in a way to inform the next step of the analysis.	12 Pts >11 to 12 Achieves Distinction All possible models are fitted, and all the relevant outputs are included in the comments made on the fitted models.	11 Pts >9 to 11 Meets expectation All possible models are fitted and there is some effort to draw inferences from the model outputs.	9 Pts >6 to 9 Pass All possible models are fitted but there is a limited amount of effort to draw inference from the model outputs.	6 Pts >3 to 6 Needs improvement Some models fitted but there is a limited amount of effort to draw inference from the model outputs.	3 Pts >0 to 3 Poor Some models fitted but there is no attempt to draw inferences from the model outputs.	0 Pts No marks Specified models are not fitted and there is no attempt to move forward with the analysis.	15 pts
Task 2: Diagnostic checking		12 Pts >11 to 12 Achieves Distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. The results are articulated in a way to demonstrate a good understanding of diagnostic checking.	11 Pts >9 to 11 Meets expectation All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	9 Pts >6 to 9 Pass Some aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	6 Pts >3 to 6 Needs improvement Some aspects of diagnostic checking are applied and most of the assumptions of applied approaches are validated.	3 Pts >0 to 3 Poor Some aspects of diagnostic checking are applied and some of the assumptions of applied approaches are validated.	0 Pts No marks There is no effort for diagnostic checking and validation of the assumptions of applied approaches.	15 pts

Criteria	Ratings							Pts
	15 Pts >14 to 15 Achieves High distinction All inferences of diagnostic checking are applied, and all assumptions of relevant outputs and given under relevant sections. A mature discussion of the presented output are used for the inferences gathered by diagnostic checking in a way to provide a mature discussion of analysis results. There is no missing section. There are no language and formatting issues.	16 Pts >14 to 16 Achieves Distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences. There is no missing section. There are no language and formatting issues.	14 Pts >12 to 14 Meets expectation All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	12 Pts >8 to 12 Pass All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	8 Pts >4 to 8 Needs improvement Inferences are addressed by relevant outputs and given under relevant sections. There are missing sections. There are minor languages and formatting issues.	4 Pts >0 to 4 Poor Inferences are not addressed by relevant outputs and given under relevant sections. Some of the output remains without being inferred. There are language and formatting issues.	0 Pts No marks Lacks the basic elements of a precise report. Some of the outputs have not been associated with inferences or there are inferences without supporting evidence.	
Task 3 Part (a): Reporting & Interpretation								20 pts
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Task 3 Part (a): R codes								

Criteria		Ratings						Pts
	dynamical way.	code chunks repeatedly.						
Task 3 Part (a): Descriptive analysis	10 Pts >8 to 10 Achieves High distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to demonstrate an outstanding understanding of concepts targeted by the descriptive statistic/plot.	8 Pts >7 to 8 Achieves Distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to shed light on further analyses.	7 Pts >6 to 7 Meets expectation Uses all suitable plots and descriptive statistics and each plot and the descriptive statistic have a suitable interpretation.	6 Pts >4 to 6 Pass Uses all suitable plots and descriptive statistics and some plots and descriptive statistic have a suitable interpretation in one of the tasks.	4 Pts >2 to 4 Needs improvement Some descriptive plots/statistics are used and/or presented all plots/statistics have interpretations.	2 Pts >0 to 2 Poor Some descriptive plots/statistics are used and/or presented plots/statistics have interpretations.	0 Pts No marks Nearly no descriptive analysis has been done and/or the descriptive analysis is insufficient.	10 pts
Task 3 Part (a): Choice of variables/model	15 Pts >12 to 15 Achieves High distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. Insightful reasons given about the selection of the model.	12 Pts >11 to 12 Achieves Distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is articulated based on the results of descriptive statistics.	11 Pts >9 to 11 Meets expectation A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is explained.	9 Pts >6 to 9 Pass A suitable model is chosen, and all variables are suitable for the aim of analysis.	6 Pts >3 to 6 Needs improvement A suitable model is chosen but some variables in a model are unsuitable for the aim of analysis.	3 Pts >0 to 3 Poor A suitable model is chosen but some or all variables in a model are unsuitable for the aim of analysis.	0 Pts No marks An unsuitable model is chosen or all variables in a model are unsuitable for the aim of analysis.	15 pts
Task 3 Part (a): Implementation of models including Forecasting	15 Pts >12 to 15 Achieves High distinction All possible models are fitted, and all the relevant outputs are clearly	12 Pts >11 to 12 Achieves Distinction All possible models are fitted, and all the relevant outputs are included in the	11 Pts >9 to 11 Meets expectation All possible models are fitted and there is some effort to draw inferences	9 Pts >6 to 9 Pass All possible models are fitted but there is a limited amount of effort to draw inference from the	6 Pts >3 to 6 Needs improvement Some models fitted but there is a limited amount of effort to draw inference	3 Pts >0 to 3 Poor Some models fitted but there is no attempt to draw inferences from the	0 Pts No marks Specified models are not fitted and there is no attempt to move forward with the analysis.	15 pts

Criteria	Ratings							Pts
	interpreted in a way to inform the next step of the analysis.	comments made on the fitted models.	from the model outputs.	model outputs.	from the model outputs.	model outputs.		
Task 3 Part (a): Diagnostic checking	15 Pts >12 to 15 Achieves High distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. A mature discussion of the information gathered by diagnostic checking is given.	12 Pts >11 to 12 Achieves Distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. The results are articulated in a way to demonstrate a good understanding of diagnostic checking.	11 Pts >9 to 11 Meets expectation All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	9 Pts >6 to 9 Pass Some aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	6 Pts >3 to 6 Needs improvement Some aspects of diagnostic checking are applied and most of the assumptions of applied approaches are validated.	3 Pts >0 to 3 Poor Some aspects of diagnostic checking are applied and some of the assumptions of applied approaches are validated.	0 Pts No marks There is no effort for diagnostic checking and validation of the assumptions of applied approaches.	15 pts
Task 3 Part (b): Intervention RBO dylm Analysis	30 Pts >20 to 30 Full marks	20 Pts >15 to 20 Good attempt	15 Pts >10 to 15 Average attempt	10 Pts >5 to 10 Partially correct	5 Pts >0 to 5 Minimally correct	0 Pts No marks		30 pts
Total points: 330								